COLMAP Experiments: Spatial AI Report

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Setup

COLMAP version: pycolmap 4.2.1. Settings: default parameters, SIFT features, exhaustive matching.

1 Part 1

What I photographed and how

Twenty pictures of a shoe were taken, lying on a black bag in an indoor environment, with a textured background characterized by a bookshelf, walls and doors. The pictures were captured using a smartphone camera, moving around the object with small rotations at constant distance. No rotation of the object was performed. The 20 images were transferred to the PC via USB cable, avoiding any loss of resolution. The set of 20 images was then subsampled to create subsets of 10, 5 and 3 images, uniformly distributed around the object.

Prediction

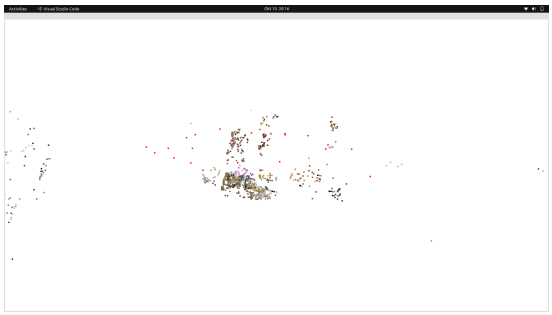
The prediction was based on the number of images used for the reconstruction. Intuitively, the more images are used, the better context the model will have, and the more points will be triangulated. I expected that the set of 20 images would result in a decent representation, unlike the 10-image set, which I expected to be partial. Lastly, both the 5- and 3-image sets were expected to be insufficient to create a model.

Image set	Registered images	3D points	Mean reproj. error (px)	Model?
20	20	3337	1.05	Yes
10	3	67	0.83	Partial
5	0	0	–	No
3	0	0	–	No

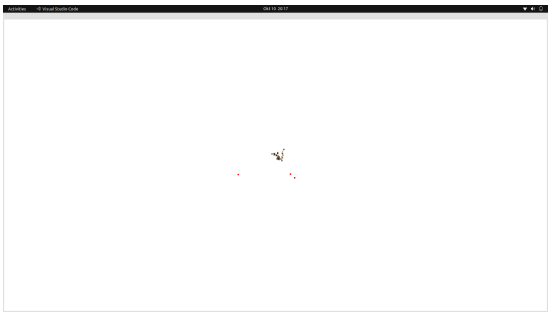
Table 1: Part 1: degradation of the reconstruction when subsampling the 20-image set.

Explanation

With 20 images, neighbouring views are close enough for SIFT to match reliably (73 of 190 image pairs verified), so COLMAP has a dense network of overlapping views to initialize from and to register further images. When subsampling, the angular distance between neighbouring views grows and the appearance of the scene changes too much for SIFT to match, so fewer pairs survive geometric verification (16/45 with 10 images, 3/10 with 5, 1/3 with 3); the average number of neighbours per image drops from 7.3 to 3.2, 1.2 and 0.7. With 10 images the model grows to only 3 images and 67 points, because new images see too few already triangulated points; with 5 and 3 images the view graph is sparse or disconnected and COLMAP finds no good initial pair.



(a) 20 images reconstruction.



(b) 10 images reconstruction.

Figure 1: Comparison between reconstructions with 20 and 10 images.

Comparison with prediction

As predicted, the 20-image set produced a complete model, even if not highly detailed. The 10-image set produced a partial model, hard to interpret. Finally, the 5- and 3-image sets failed to produce any model, as expected. Focusing on the 10-image set, only 3 images were registered because the remaining views share too few matches with the model, and the reprojection error is lower (0.83 px against 1.05 px for the 20 images), but this only reflects the few points left, not a better model.

Note on preliminary attempts

Before the final 20-image set, two preliminary attempts were made. A 19-image set registered only 4 images and produced 230 3D points. An earlier set produced 1553 points. Both were worse than the final 20-image set (20 registered images, 3337 points, 1.05 px). This confirms that the distribution and quality of the views matter more than the sheer number of images.

2 Part 2

What I photographed and how

For the second experiment, I took 10 pictures of an indoor environment, similar to a kitchen. No translation was performed: the camera was kept as still as possible, performing only rotations.

Prediction

I expected feature extraction and matching to succeed (strong overlap, textured scene), but the pipeline to fail at the initialization stage: with a pure rotation the views are related by a homography and there is no baseline, so no initial pair with a sufficient triangulation angle exists and no sparse model can be created.

Results

As predicted, matching succeeded (all 10 images have verified matches) but initial pair selection failed for every candidate pair: 0 registered images, 0 3D points.

Explanation

The failure comes from the geometry of the motion: the camera rotated in place without translating, so there is no parallax and depth cannot be observed. All rays share the same optical center, and the images are related by a homography, not by epipolar geometry with a baseline. Blur and exposure are not the causes.

3 Part 3

What I did

Three images with lateral translation were added to provide the missing baseline, reaching a total of 13 images. No settings were changed.

Prediction

The initialization was expected to succeed, thanks to the baseline provided by the three new images. However, a complete model was not expected, since the majority of the images still have no parallax and cannot be triangulated.

Results

COLMAP processed 13 images and registered only a subset of them. As expected, the model is partial. The final model contains 3 registered images (03.jpg, 06.jpg, 14.jpg), 448 3D points, and a mean reprojection error of 0.85 px, which is low but does not by itself prove correctness (Figure 2).

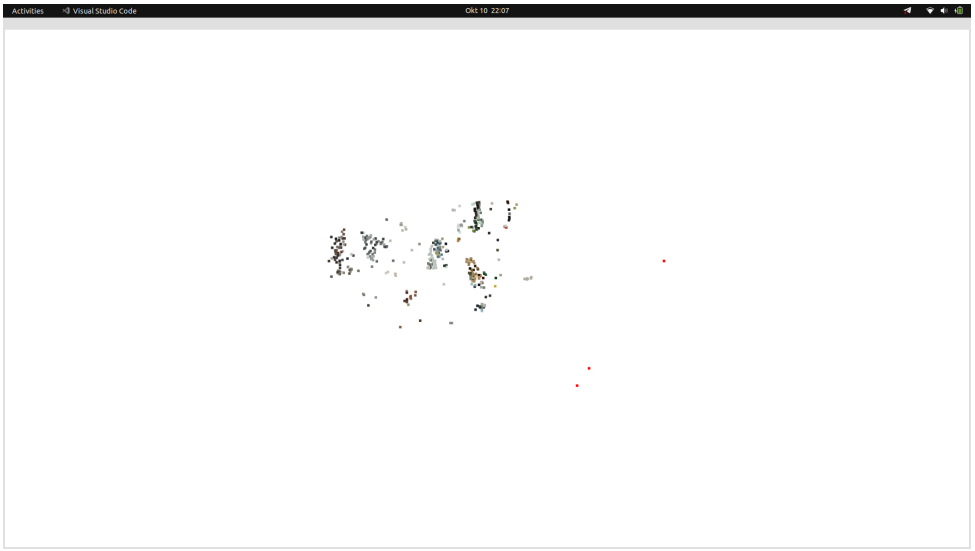


Figure 2: Part 3: Results of the repair.

Explanation and correctness

The baseline introduced by one of the new images (14.jpg) allows initialization together with 06.jpg, and a third image (03.jpg) is then registered. The remaining 10 images (8 pure-rotation images and 2 of the new ones) could not be registered: the rotation-only images add no parallax, and the two other translated images probably did not share enough matches with the registered views. The model reprojects well on the three registered views, but a low reprojection error does not prove correctness: with such a short baseline depth is poorly constrained, so points may be wrong along the viewing direction even if they reproject well. It is also not a complete reconstruction of the scene. This shows that a model that exists is not necessarily a correct or complete model.



(a) 20 images reconstruction image example.



(b) 2nd experiment image example.

Figure 3: Example of images for the 2 experiments.